

MA 2320 Final Exam Sample 1

Name: _____

No calculators, no phones, no electronics allowed. Must show work for credit.

1. Compute, using the definition, the exact value of

$$\int_1^4 x^2 dx$$

2. Compute the area between the graphs of $y = x^2$, and $y = 4 - x^2$.
3. Compute the volume of revolution when the region between the graphs of $y = x$, and $y = \ln(x)$ from $y = 0$ to $y = 1$ is revolved about the y-axis using either discs or shells.
4. Compute the volume of revolution when the region between the graphs of $y = 0$, and $y = e^{x^2}$ is revolved about the y-axis using either discs or shells.
5. $\int e^{x^3-1} x^2 dx$
6. $\int x^2 e^{2x} dx$
7. $\int \sin^2(4x) dx$
8. $\int \frac{1}{\sqrt{1-4x^2}} dx$
9. $\int \frac{1}{x(x^2+1)} dx$
10. For the following determine if the series converges or diverges. For all of these, as in class, show the following: 1. test/series type, 2. Conclusion 3. Criteria 4. work.

(a) $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}}{2n^2+1}$

(b) $\sum_{n=1}^{\infty} \frac{1}{n^3}$

(c) $\sum_{n=1}^{\infty} \frac{2^n}{n^n}$

$$(d) \sum_{n=1}^{\infty} \left[\frac{2\sqrt{n}+1}{\sqrt{3n+1}} \right]^n$$

11. Find the interval of convergence and the radius of convergence for

$$\sum_{n=1}^{\infty} \frac{x^n}{n2^n}$$

12. Find the Taylor/Maclaurin series from the definition for $f(x) = \sin 5x$.

13. Find the Taylor/Maclaurin series from a known series for $f(x) = x \sin x^2 - x^3$.

14. Graph $x^2 - y^2/4 = 4$.

15. Graph $r(\theta) = 2 \cos(\theta)$.

16. Translate into a parametric equation.

(a) $y = 3x + 2$.

(b) $x = 3y^2 + 2$.

(c) $x^2 + y^2 = 1$.